

SCN UNIVERSITY

Group 14

Smart Mobile Application Development

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ISoftware Design Report

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1 Introduction

1. Project Objectives and Summary

This project aims to design and develop a smart social application named "LinkUp" to address the core challenges faced by contemporary young people in social interactions and career development. Current social platforms commonly suffer from issues such as excessive utilitarianism, superficial connections, and the disconnection between online and offline scenarios. To tackle these problems, LinkUp innovatively proposes an O2O (Online-to-Offline) closed-loop social model characterized by "online precise matching + offline scenario-based activities + online relationship consolidation." Targeting university students, young professionals, and startups as its core user groups, and driven by AI technology with a "Citrus Culture" as its entry point, the product strives to build an efficient, valuable, and warm platform for connection and collaboration. Its goal is to transform every social connection from a "chance encounter" into "inevitable value."

2. Project Scope

The scope of this project encompasses the entire process from conceptual design to the implementation of core functionalities for the LinkUp application, specifically including:

- **Market and User Analysis:** Conduct an in-depth analysis of the core pain points and market demands of the target user groups.
- **Product Solution Design:** Define the product solution centered on the O2O social closed-loop, detailing the functional modules and user experience for the three key stages: online intelligent matching, offline scenario-based connections, and online relationship consolidation.
- **Technical Architecture Planning:** Specify the key technologies required to implement the product's functionalities, particularly the AI-powered intelligent matching and recommendation algorithms.
- **Business Model Outlook:** Plan the long-term revenue path centered on a "Freemium" model.

This report will systematically elaborate on the requirements, overall design, key technologies, and business model of the LinkUp project based on the above scope.

2 Requirements Specification

The internship took place in the months of June to September. Most of the work was done either at the office of LoremIpsum, Location, or my place of residence when working from home. The first two weeks were spent

reading the literature and discussing possible approaches. The next X weeks were spent on dolor sir amet. Finally the model was evaluated with consectetur adipiscing elit in the last Y weeks.

1. User Requirements

This section translates the target user needs of LinkUp into specific, actionable functional requirements.

1. User Requirements

After conducting an in-depth analysis of market pain points, we have identified the following core user requirements:

- **University Students:** Their core dilemma is "having no guidance" at the beginning of their careers. Their need is not only to access recruitment information but, more importantly, to obtain quality internal referral opportunities and channels for in-depth communication with industry practitioners. This helps break down information barriers, transform superficial industry knowledge into clear career planning, and ultimately address the fundamental confusion regarding internship and employment directions.
- **Young Professionals (1–3 years of experience):** The core demand of this group is to break the "fighting alone" state in their career development. They strongly need to break out of their current closed social circles, connect with people from different fields (e.g., investment, marketing, technology) at a low cost, and gain cross-border resources and perspectives. Simultaneously, they are dissatisfied with theoretical learning and urgently need practical collaboration scenarios based on real projects. By working with others to complete specific tasks, they can tangibly enhance their skills and achieve the leap from "knowing" to "doing."
- **Startups:** Their core need is to solve the "stretched thin" problem under resource constraints. Regarding talent acquisition, they require precise and cost-controlled recruitment solutions to efficiently screen like-minded individuals who possess both the necessary skills and an entrepreneurial spirit. For business expansion, they urgently need efficient and credible cross-border resource connection channels to access key resources such as supply chains, channels, and investment/financing, avoiding repeated setbacks due to a lack of trust endorsement.

Common Needs of All Users: Beyond the specific pain points above, contemporary young users generally expect social activities to shift from being "utilitarian" to "value-driven," desiring to establish deep connections that bring substantial growth and collaboration. Additionally, the seamless integration of online and offline scenarios (OMO) has become a rigid requirement. Users expect to form a complete closed loop of "online precise matching, offline high-quality interaction, and online effective consolidation," avoiding issues such as online relationships becoming meaningless or offline activities being one-time events.

2. Functional Requirements

To fulfill the above user requirements, the application must include the following core functionalities:

- **Intelligent Matching and Recommendation System:** The system should perform AI-based matching using multi-dimensional data (e.g., skill complementarity, career goals, activity preferences), proactively recommending "high-value connections" and "suitable activities" to users, and display the matching rationale. After users sign up for an activity, the system should recommend fellow participants for pre-event communication.

- **Scenario-Based Activity Management Platform:** The platform must support the creation, discovery, and registration of various offline activities (e.g., industry salons, skill workshops, networking events) and integrate on-site interaction tools such as activity check-in and AR business card scanning to facilitate ice-breaking.
- **Relationship Consolidation and Activation Mechanism:** The system must automatically convert offline interactions (e.g., exchanging business cards, completing collaborations) into online profile records (e.g., acquaintance tags, project achievements). Simultaneously, by employing AI-powered reminders for follow-up interactions and providing online collaboration spaces to reuse offline project groups, it ensures the sustained value of connections.
- **"Buddy" Culture Light Social System:** To address lightweight social needs, provide a fast and accurate "buddy" matching function (e.g., for dining, sports) and establish a bidirectional evaluation system post-activity to optimize the matching algorithm and incentivize high-quality social interactions.

3 Overall Design

The overall design of this project revolves around the O2O closed-loop social model of "online precise matching, offline scenario-based connection, and online relationship consolidation." The application's core architecture aims to translate the concept of "buddy culture" into a smooth and efficient user experience through a clear user journey.

1. Core Architecture and Functional Workflow

- **User Entry and Discovery (Homepage Card Stream):** The application homepage adopts a card stream format to present activity information. It prominently displays key elements such as "time, activity theme, and participant avatars," supporting users in completing activity discovery and decision-making within seconds through minimalist swipe gestures: left to bookmark and right to register.
- **Intelligent Matching and Ice-breaking (Matching Analysis System):** When two parties show interest in the same activity, the system automatically generates a four-dimensional radar chart based on shared interests, availability, industry, and activity frequency. This visual tool quantifies compatibility and provides one-click ice-breaking topic suggestions, significantly lowering the psychological barrier and operational cost of initial communication.
- **Scenario-Based Connection and Interaction (Real-time Activity Hub):** During an activity, a dedicated activity hub interface displays key information (e.g., time, location) and integrates tools such as AI-powered real-time recommendations for fellow participants and AR business card exchange. This ensures that offline interactions are efficient and focused.
- **Relationship Consolidation and Value Reuse (Modular Personal Space):** A user's personal homepage is no longer a static resume but is composed of freely assemblable modules such as "Tag Cards," "Achievement Cards," and "Activity Cards." Offline interactions (e.g., exchanging business cards, completing collaborations) are automatically transformed into online tags, achievements, and activity updates. This enables seamless, visual consolidation of relationships and provides a foundation for subsequent connections.

2. Core Interaction Philosophy

The overall design adheres to a low-pressure social philosophy of **"instant matching, immediate meetup, and natural parting."** Through a minimalist information architecture, visual matching feedback, and contextually integrated tools, the user experience loop—from online screening to offline meetings and then to online consolidation—is kept lightweight and smooth. This approach precisely aligns with the dual desires of young users for freedom and efficiency.

4 User Interface Design

The user interface design serves as the concrete implementation of the overall design philosophy. Its core objective is to balance professionalism and friendliness, while guiding user behavior through visual language and conveying the emotional value of **"stress-free connection."**

- **Color System:** Blue is adopted as the primary color, establishing a professional and trustworthy platform tone. High-saturation orange and green are used as secondary and interactive colors, injecting vibrancy and approachability. Extensive white space is utilized to create a clear, non-oppressive visual feel. The overall color scheme aims to evoke positive associations with an urban, trendy lifestyle.
- **Visual Emotion:** The design successfully achieves a progression from function to emotion. A structured layout and data visualization (e.g., radar charts, progress rings) establish a professional image. Simultaneously, high-saturation buttons, playful badges, and achievement systems provide positive reinforcement, making the interface emotionally warm while remaining efficient.

2. Key Interface and Component Innovations

- **Homepage Card Stream:** Employing a card stream interaction model similar to short-video platforms, it minimizes user browsing and decision-making costs, enabling **"one-second activity selection."**
- **Modular Personal Information Cards:** Traditional lengthy personal profiles are deconstructed into three freely sortable modules: **"Tag Cards"** (icon + keywords), **"Achievement Cards"** (visualized progress), and **"Activity Cards"** (social feed-style updates). This allows users to assemble personalized homepages, significantly reducing information reading load.
- **Matching Radar Chart and Ice-breaking Tools:** Upon successful matching, a compatibility radar chart is presented immediately, accompanied by conversational conclusions and pre-set ice-breaking topic buttons. This provides users with instant, effective support for breaking the ice, transforming matches into meaningful conversations.
- **Integrated Chat Interface and Activity Status:** Key activity status is pinned within the chat window, ensuring communication consistently revolves around offline meetings. This strengthens the product's action-oriented nature and completes the experience loop.

5 Key Technologies

During the development process of this project, multiple mobile development technologies were integrated, and specific technical challenges were addressed to meet the core product features.

1. Main Technology Stack

- **Development Platform and Language:** Based on the Android Studio integrated development environment, Kotlin is adopted as the primary development language, leveraging its modern language features to enhance development efficiency and code security.
- **Application Architecture:** Following the MVVM design pattern and utilizing the Jetpack component library (e.g., ViewModel for managing UI-related data, LiveData for data change observation and responsive UI updates), the architecture ensures good code structure and maintainability.
- **UI Implementation:** Strictly adhering to the Material Design specifications, interfaces are laid out using XML to provide users with a mainstream, aesthetically pleasing, and consistent operational experience.
- **Core Feature Technologies:**
 - **Intelligent Matching:** Implemented a matching algorithm based on multi-dimensional data such as user behavior, interest tags, and career goals. A lightweight version is deployed on the Android client to enable personalized recommendations for "connections" and "buddies."
 - **Real-time Interaction:** Developed a real-time chat feature that supports instant transmission of multimedia messages including text and images. It also enables social interactions such as likes and comments on community feeds.
 - **Data Management:** Utilized the SQLite database for local data persistence and designed corresponding data synchronization mechanisms to support offline use of certain features and ensure data consistency.

Technical Challenges and Solutions

- **Lightweight Deployment of AI Algorithm on Android**

Balancing the accuracy of the matching algorithm with runtime efficiency under the limited computing power of mobile devices is a significant challenge.

Solution: Optimized and pruned the matching model, prioritized a strategy where complex computations are completed on the server-side and lightweight inference is performed on the client-side. Leveraged Kotlin Coroutines for asynchronous task handling to avoid blocking the main thread and ensure UI fluidity.
- **Fragmentation Adaptation Across Android Devices**

Supporting multiple brands (e.g., Huawei, Honor, OnePlus) and different Android versions presents significant compatibility issues.

Solution: Conducted extensive compatibility testing on real devices across various brands and system versions. Performed targeted adaptation adjustments for specific device differences (e.g., HarmonyOS) to ensure stable operation of core features on mainstream models.
- **Real-time Synchronization of O2O Data Loop**

Ensuring that offline activities (e.g., event check-ins, AR business card exchanges) are synchronized to online profiles in real-time and accurately, particularly in weak network environments, is a major challenge.

Solution: Implemented robust network request retry mechanisms and local caching queues to automatically synchronize data once the network is restored. Applied transaction locks for critical data operations (e.g., collaboration invitations) to ensure data consistency.

• **Real-time Interaction in the "Buddy" Activity Module**

Implementing rapid team formation for small group activities (2-6 people), real-time message delivery, and progress synchronization requires overcoming limitations in Android background task management and push notification mechanisms.

Solution: Appropriately utilized system-level push services and long-connection technologies, optimized message push strategies, and employed foreground services during active sessions to maintain connection stability as much as possible, thereby enhancing the real-time communication experience.

6 **Testing and User Experience Analysis**

To ensure application quality, this project underwent rigorous third-party testing and collected direct feedback from target users.

1. **Third-party Cloud Platform Test Results**

The project commissioned a third-party cloud testing platform to conduct automated multi-device testing on the application package. Key performance indicators were as follows:

已测试终端列表

问题类型	安装失败	启动失败	运行失败	卸载失败	Monkey失败		
品牌	全部	一加	华为	荣耀			
系统版本	全部	11	13	3.0.0	4.2.0	7.0	
屏幕尺寸	全部	5.0	6.1	6.5	6.53	6.55	6.7
内存	全部	3-4GB	4-6GB	6-8GB			

设备名称	状态	初始化	安装	启动	遍历	monkey	卸载	Crash	ANR	Exception	错误描述	操作
华为 nova	失败	✓	✗	✗	-	-	-	0	0	0	安装失败	补测
一加 8T	通过	✓	✓	✓	✓	-	✓	0	0	0	--	补测
华为 Mate 30 Pro	通过	✓	✓	✓	✓	-	✓	0	0	0	--	补测
荣耀30青春版 HarmonyOS	通过	✓	✓	✓	✓	-	✓	0	0	0	--	补测
华为 P40	通过	✓	✓	✓	✓	-	✓	0	0	0	--	补测
华为 Mate 40E	通过	✓	✓	✓	✓	-	✓	0	0	0	--	补测
一加 9 Pro	通过	✓	✓	✓	✓	-	✓	0	0	0	--	补测
华为 Mate 30 Pro	通过	✓	✓	✓	✓	-	✓	0	0	0	--	补测

Figure 1: Performance test results from third-party cloud platform

Results Assessment: The test results indicate that the application performs well on most mainstream devices. The average startup time is generally below 1.5 seconds, with CPU and memory usage at reasonable levels, confirming overall operational smoothness through data verification. Although the installation time on the OnePlus 9 Pro is relatively high (potentially requiring further package optimization), post-startup performance is excellent.

2. **User Experience Evaluation and Feedback**

Field testing was conducted with 15 target users (university students aged 18-22), and feedback was collected through questionnaires and interviews.

- **Positive Feedback (Core Highlights):**
 - **AI Matching Accuracy:** Users recognized the relevance of the system-recommended "connections" and "buddies."

- **Buddy Activity Format:** The lightweight, goal-oriented "buddy" model was highly welcomed and considered effective in reducing social pressure.
- **Operational Smoothness:** Subjective experience aligned with test data; users found the application operation smooth.

- **Negative Feedback and Improvement Analysis:**

- **Functional Logic Issues:** The most prominent problem was "no automatic redirection after successful registration" (affecting 40% of users). Other issues included "AI matching unresponsive" and "activity check-in status not synchronizing." These problems exposed flaws in interface callbacks and data synchronization logic, which have been prioritized for fixing.
- **User Experience Issues:**
 - * **Cumbersome Registration Process:** Users considered the number of required fields excessive. **Improvement:** Simplify the process and add third-party one-click login.
 - * **Insufficient Ice-breaking Guidance for Activities:** Users were unsure how to start a conversation after matching. **Improvement:** Automatically push ice-breaking topics and create temporary chat groups after activity registration.
 - * **AR Feature Lag on Older Devices:** Compatibility needs enhancement. **Improvement:** Reduce AR rendering resolution for lower-performance devices.
 - * **Confusing Interface Hierarchy:** Some feature entry points were unclear. **Improvement:** Restructure the homepage layout to highlight core functions and optimize visual information hierarchy.

Summary: User testing validated market acceptance of the core product concept (AI matching + buddy culture). The exposed issues primarily centered on program stability and interaction details. Targeted improvements can significantly enhance the completeness and satisfaction of the user experience.

7 Conclusion and Outlook

This report has elaborated on the complete process of the LinkUp smart social application, from market insights and product design to technical implementation and testing optimization. The project successfully integrated "buddy culture" with professional networking needs, innovatively proposed an O2O social closed-loop model that blends online and offline interactions, and developed a functional product prototype with core features.

1. Project Achievements Summary

The project successfully delivered a fully functional Android application prototype, implementing four core modules: "Connection Recommendation and Matching," "Real-time Chat and Community Interaction," "Personal Profile Editing," and "Offline Activity Push and Registration." By introducing innovative interactions such as AI-powered intelligent matching algorithms, a lightweight buddy social model, and AR business cards, the project effectively validated the feasibility of addressing young people's social and career development pain points through technological means. Third-party testing and user field tests demonstrated that the application runs smoothly on mainstream devices, and its core concepts received positive recognition from target users.

2. Development Challenges and Value

During the development process, the team successfully tackled a series of technical challenges, including lightweight deployment of AI algorithms, fragmentation adaptation across Android devices, and real-time synchronization of online-offline data. These practices not only ensured the technical feasibility of the product but also accumulated valuable experience for the development of similar applications. Issues related to program logic and interaction details revealed during user testing have been addressed with effective improvement plans. The implementation of these fixes and optimizations will significantly enhance product stability and user experience.

3. Future Outlook

Looking ahead, LinkUp has vast development potential. First, based on user feedback, priority should be given to enhancing the user experience by simplifying the registration process, strengthening ice-breaking guidance, and optimizing the interface hierarchy. Second, as the user base grows, the AI algorithm can be further refined by incorporating richer user behavior data to make matching and recommendations more accurate. Additionally, exploring diverse activity scenarios (e.g., online collaborative projects, industry flash events) can enrich social value. In terms of the business model, the "Freemium" model can be gradually implemented, offering value-added services such as targeted recruitment and branded events for B2B clients. Ultimately, LinkUp aims to become a trusted ecosystem platform for Generation Z professionals, integrating networking, skill enhancement, and value creation.